lab09

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Analysis of Candy Data

```r
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
```

```
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

Q1. How many different candy types are in this dataset?

```r
dim(candy)
```

```
[1] 85 12
```

There are 85 different candy types.

Q2. How many fruity candy types are in the dataset?

```r
table(candy$fruity)
```

```
 0  1
47 38
```

There are 38 fruity type candies in the data set.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy are M&Ms:

```r
candy["M&M", ]$winpercent
```

```
[1] 66.57458
```

M&Ms win percent was 66.57%

Q4. What is the winpercent value for "Kit Kat"?

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Kit kats have a win percent of 76.77%

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```r
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Tootsie Roll Snack Bars had a win percent of 49.65%

#Using the Skim library to summarize candy

```r
library(skimr)
```

```
Warning: package 'skimr' was built under R version 4.4.3
```

```r
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Win percent is on a scale of around 50 while the rest of the variables are around 0.5.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

The 0 and 1s represent presence or absence with a 0 representing the candy does not contain the variable and a 1 representing it does.

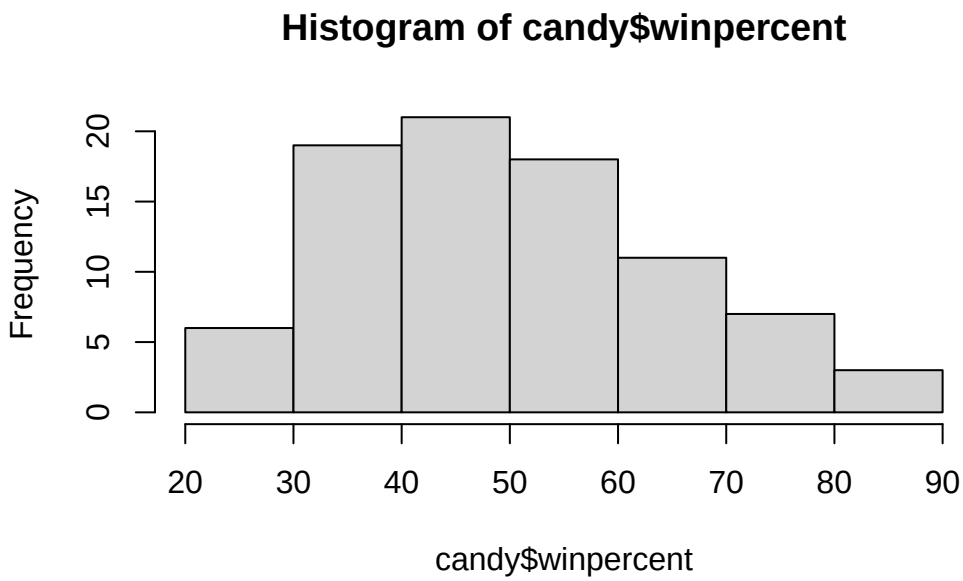
Exploritory Analysis

Analyzing the candy data set to visualize relationships between candies. >Q8. Plot a histogram of winpercent values using both base R an ggplot2.

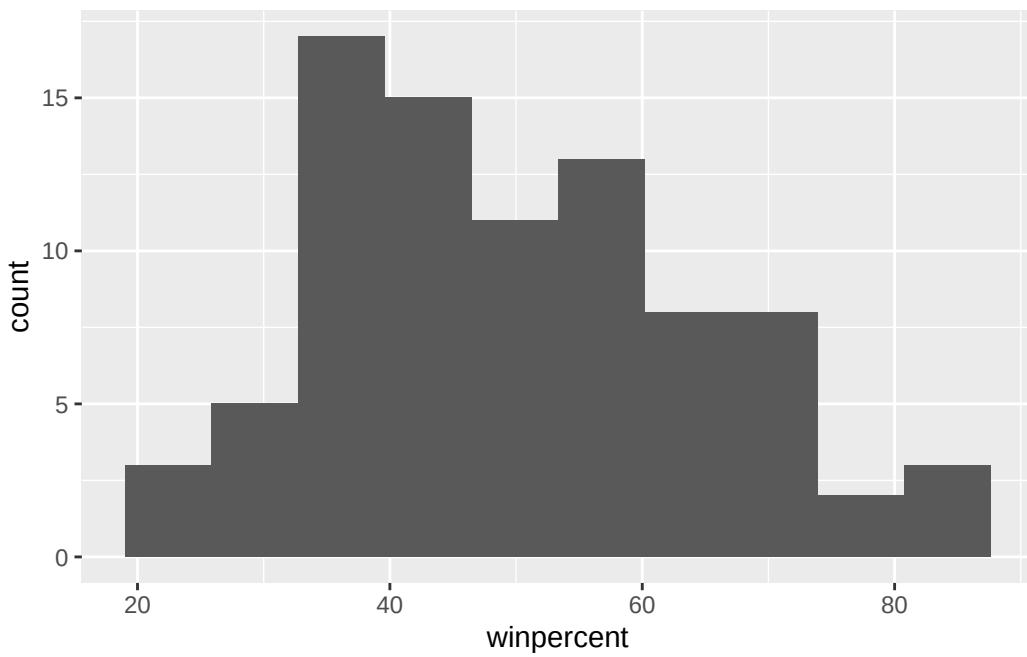
```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
hist(candy$winpercent)
```




```
ggplot(candy, aes(winpercent)) +
  geom_histogram(bins = 10)
```



Q9 Is the distribution of winpercent values symmetrical?

the distribution of win percents are slightly skewed right towards candies with a high win percent.

Q10. Is the center of the distribution above or below 50%?

```
median(candy$winpercent)
```

```
[1] 47.82975
```

The center of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
chocolate <- candy %>%
  filter(chocolate == 1)
fruity <- candy %>%
  filter(fruity == 1)
mean(chocolate$winpercent)
```

```
[1] 60.92153
```

```
mean(fruity$winpercent)
```

```
[1] 44.11974
```

On average, the chocolate candy had a win percent of 60.92% while the fruity candy had a win percent of 44.12%, so on average chocolate candy was ranked higher.

Q12. Is this difference statistically significant?

```
t.test(chocolate$winpercent, fruity$winpercent)
```

```
    Welch Two Sample t-test

data:  chocolate$winpercent and fruity$winpercent
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Since the p value for the t test comparison was 2.87e-8 which is less than 0.05, the difference between chocolate candy and fruity candy win rates are statistically significant.

Overall Candy Ranking

Ordering the data set by win percent

Q13. What are the five least liked candy types in this set?

```
ordered <- candy %>%
    arrange(winpercent)
head(ordered, 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
```

```
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

The least liked candies were Nik L Nips, then Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```r
tail(ordered, 5)
```

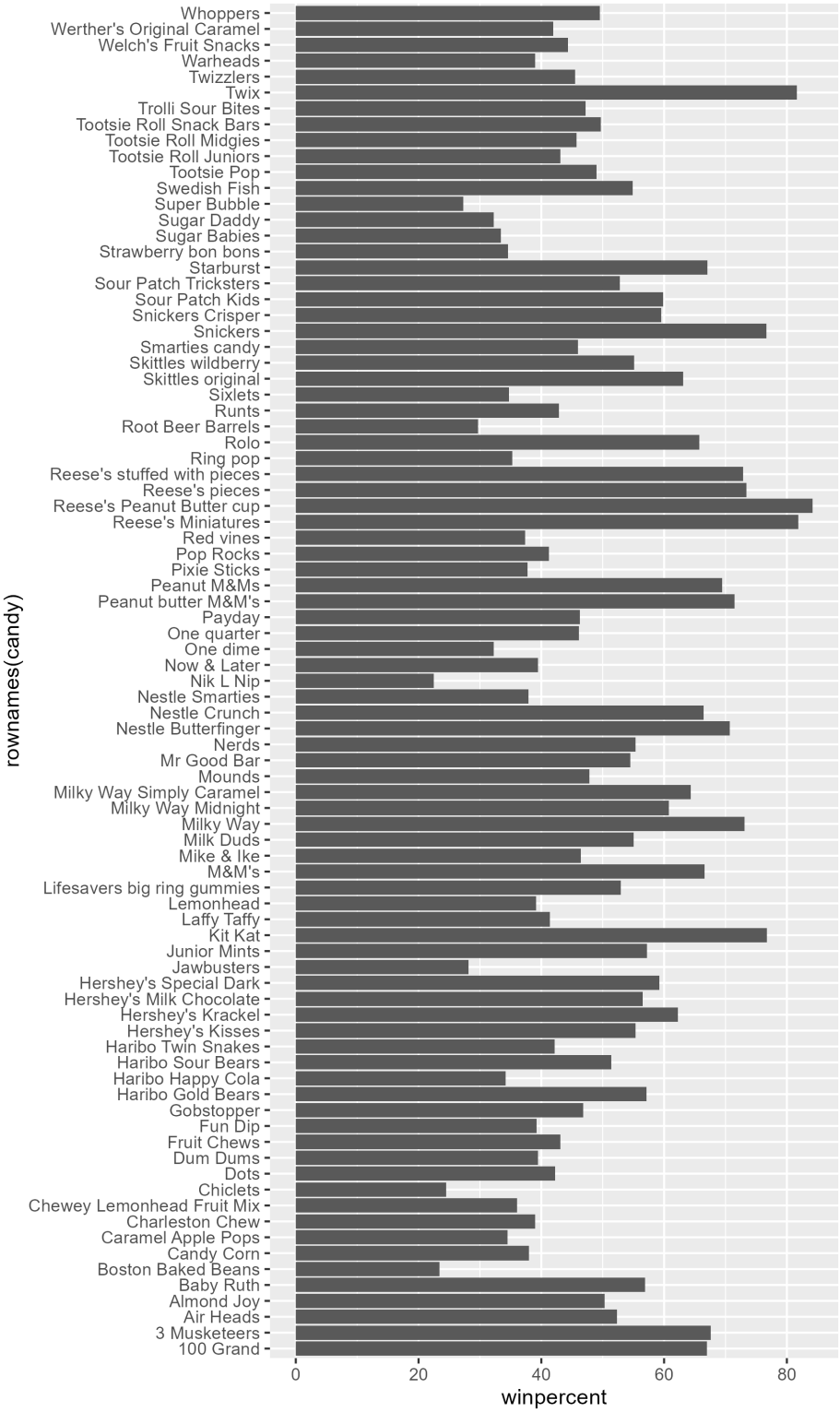
```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                  0    0   1        0        0.546
Kit Kat                                   1    0   1        0        0.313
Twix                                      1    0   1        0        0.546
Reese's Miniatures                        0    0   0        0        0.034
Reese's Peanut Butter cup                 0    0   0        0        0.720
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

The most liked candies were Reese's Peanut Butter cup, then Reese's Miniatures, Twix, Kit Kats, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy, aes(winpercent, rownames(candy)))+
  geom_col()
```

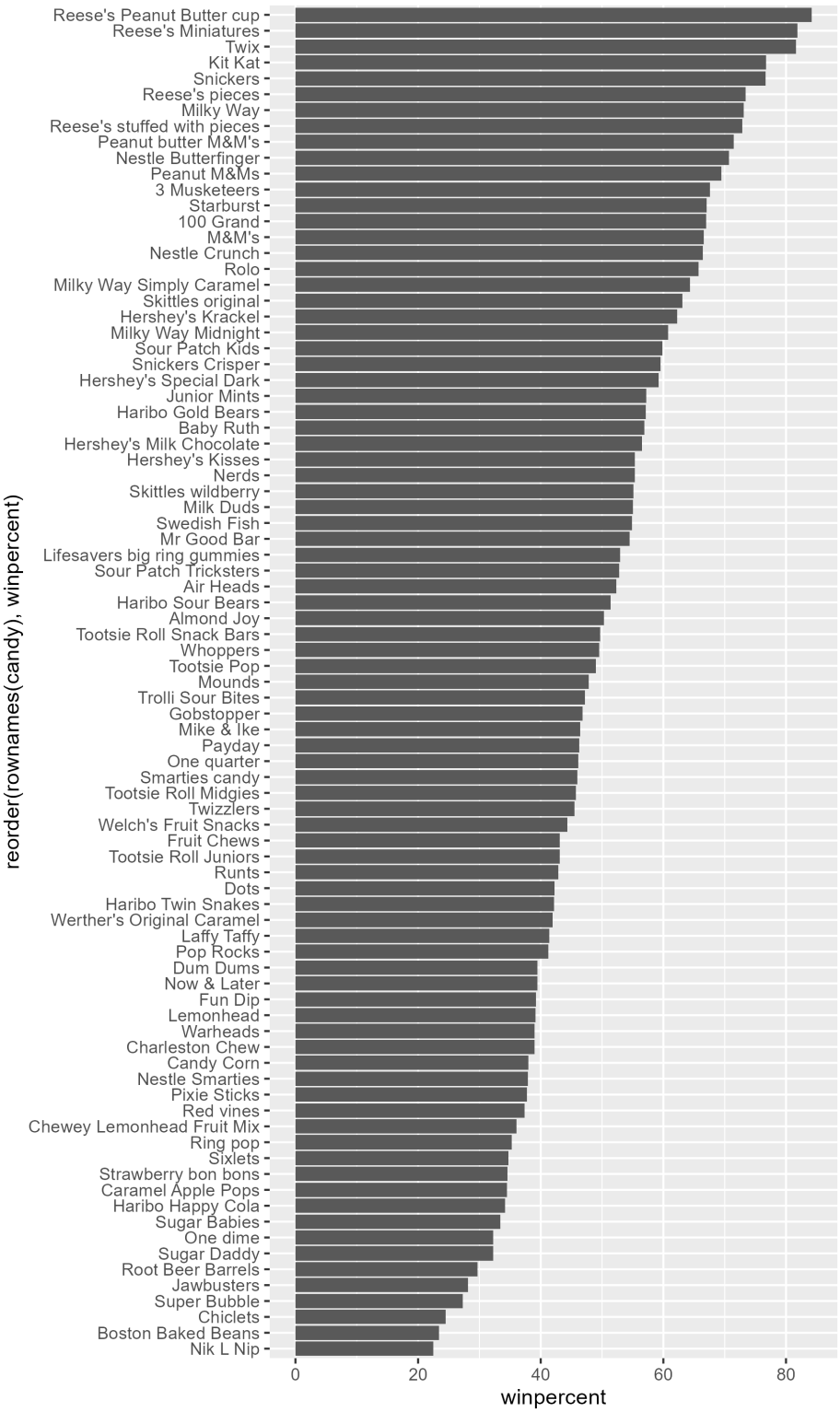
```r
  ggsave("barplot1.png", height = 10, width = 6)
```


Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy, aes(winpercent, reorder(rownames(candy), winpercent))) +
  geom_col()
```

```
 ggsave("barplot2.png", height = 10, width = 6)
```



Adding color to

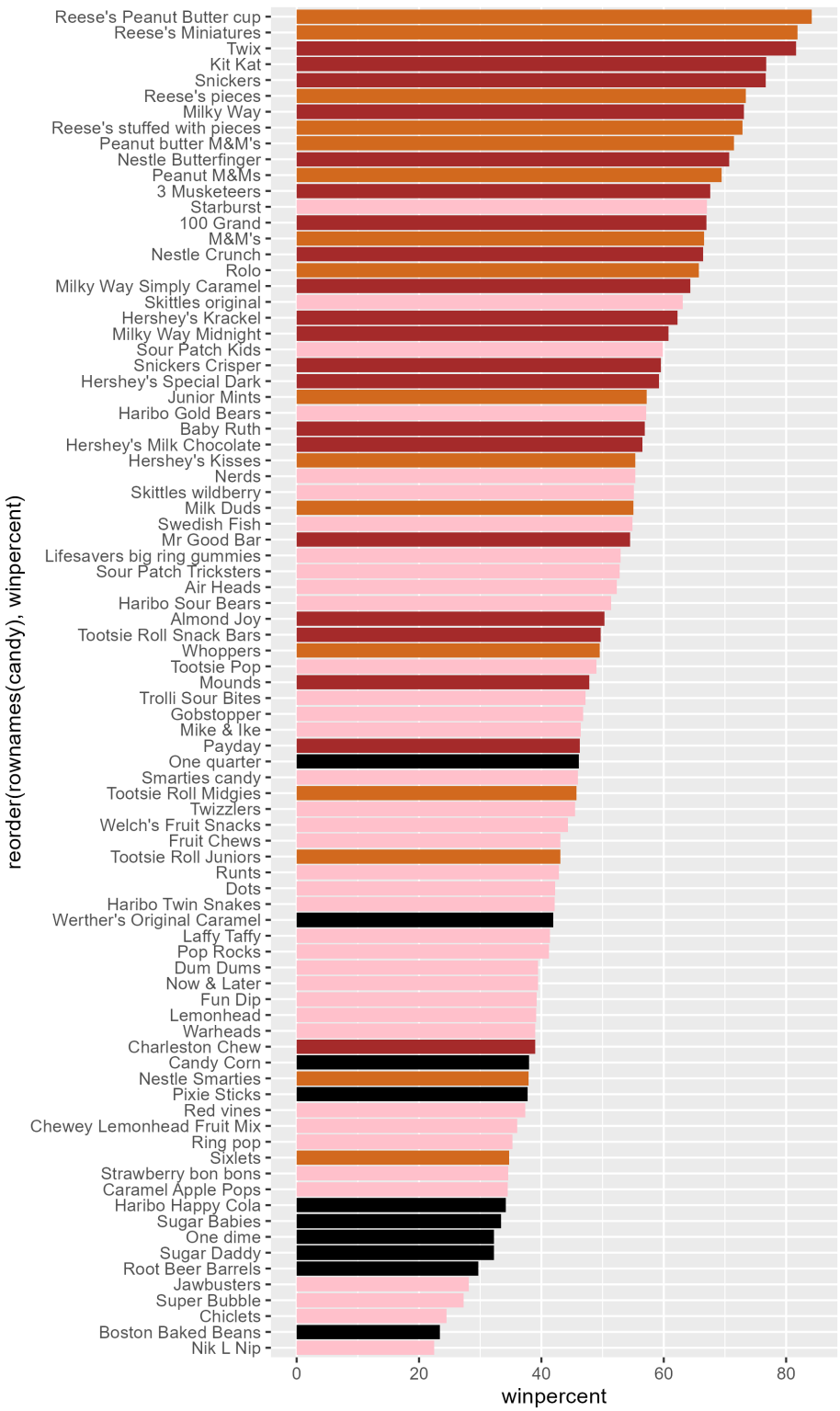
the graph

Setting color values for each candy type

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```

```
ggsave("barplot3.png", height = 10, width = 6)
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

From the graph the worst ranking candy were Nik L Nips.

Q18. What is the best ranked fruity candy?

From the graph the best ranked fruity candy were Starbursts.

Comparing Win Percent and Price

```
library(ggrepel)
```

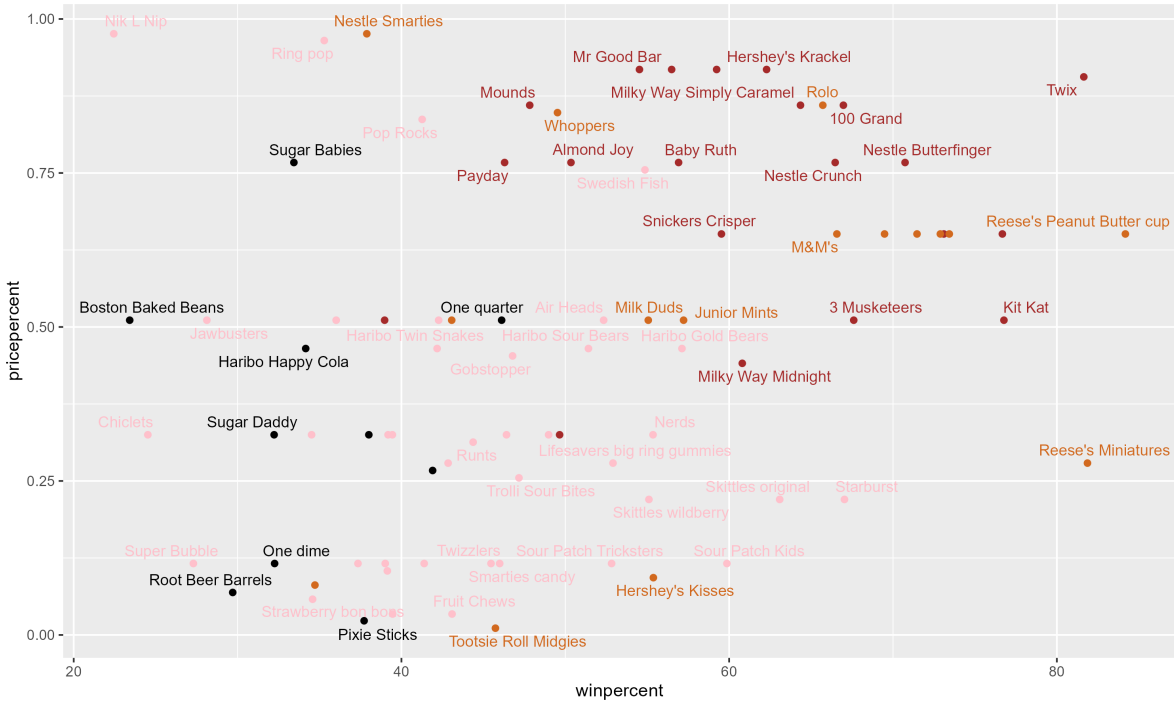
```
Warning: package 'ggrepel' was built under R version 4.4.3
```

```
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
ggsave("dotplot.png", height = 6, width = 10)
```

```
Warning: ggrepel: 27 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures have among the highest win percent while having a price percentile of around 25%.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
head(candy %>%
  arrange(desc(pricepercent)))
```

```
                        chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                       0      1       0              0      0
Nestle Smarties                 1      0       0              0      0
Ring pop                        0      1       0              0      0
Hershey's Krackel               1      0       0              0      0
Hershey's Milk Chocolate        1      0       0              0      0
Hershey's Special Dark          1      0       0              0      0
                        crispedricewafer hard bar pluribus sugarpercent
Nik L Nip                               0    0   0        1        0.197
Nestle Smarties                         0    0   0        1        0.267
```

```
Ring pop                                0    1   0        0        0.732
Hershey's Krackel                       1    0   1        0        0.430
Hershey's Milk Chocolate                0    0   1        0        0.430
Hershey's Special Dark                  0    0   1        0        0.430
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
Hershey's Special Dark          0.918   59.23612
```

The top 5 most expensive candies are Nik L Nips, Nestle Smarties, Ring Pops, Hershey's Krackels, and Hershey's Milk Chocolate. The least popular most expensive candy were Nik L Nips.

Exploring the Corelation Strucutre

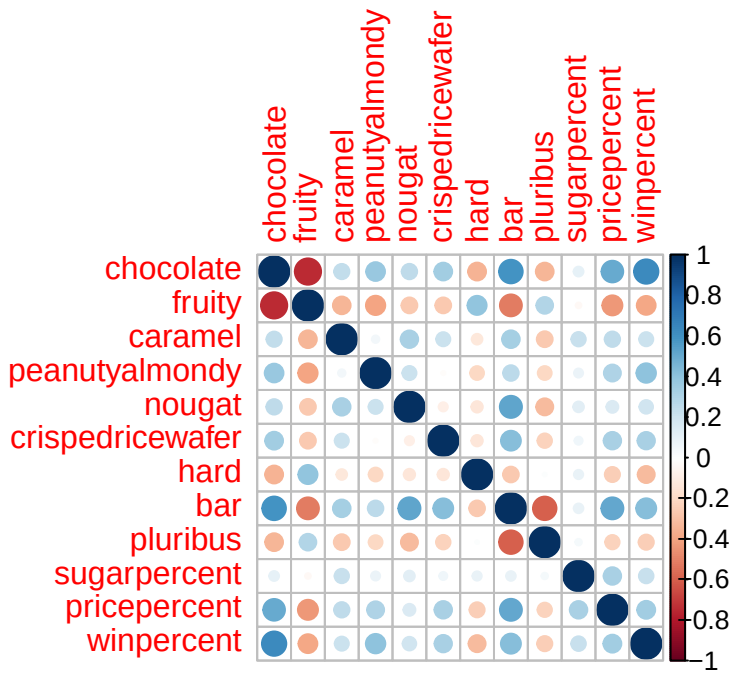
Creating a correlation matrix of the candy types using the `corrplot` package.

```r
library(corrplot)
```

```
Warning: package 'corrplot' was built under R version 4.4.3
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The variables with the most anti correlation are chocolate and fruity and bar and pluribus. This means that it is unlikely that any candy will be both chocolate and fruity or a bar and in a pluribus packaging.

Q23. Similarly, what two variables are most positively correlated?

The most positively correlated variables are chocolate and winpercent.

Principal Component Analysis

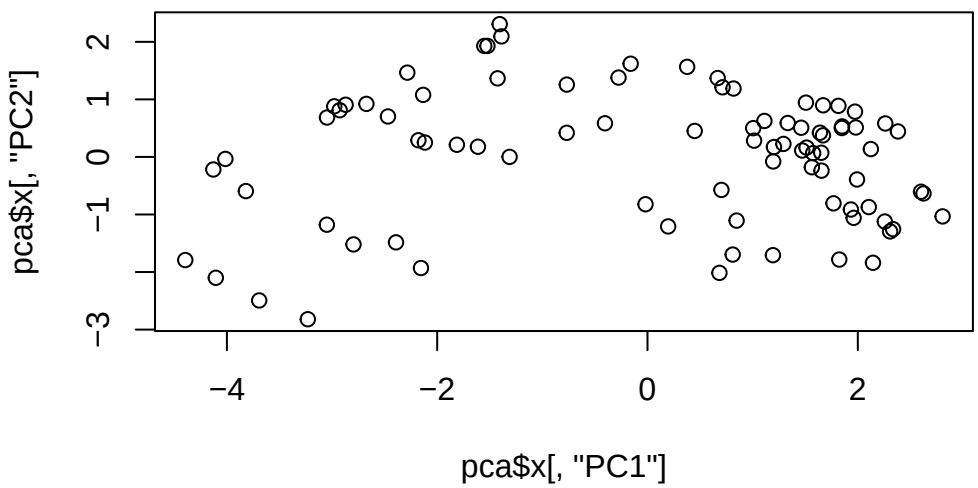
```
pca <- prcomp(candy, scale = T)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
```

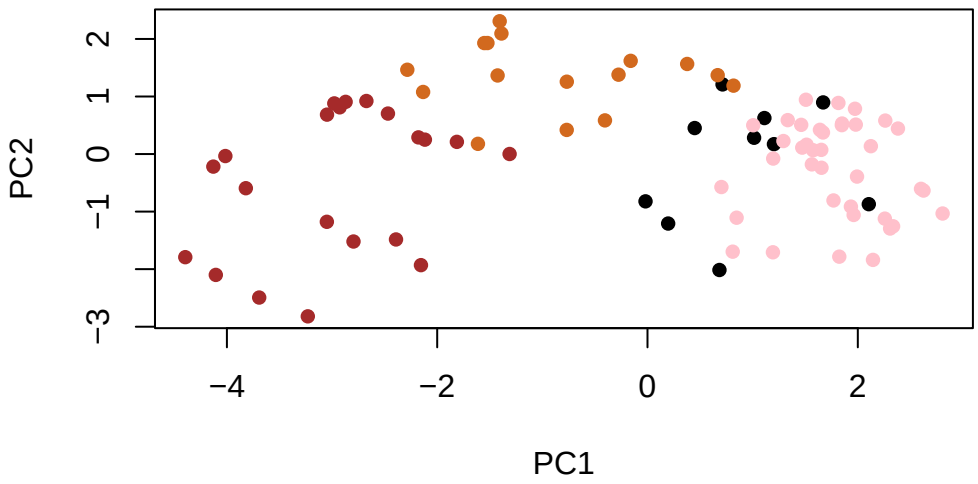
```
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

Plotting PC1 and PC2

```r
plot(pca$x[, "PC1"], pca$x[, "PC2"])
```




```r
plot(pca$x[,1:2], col=my_cols, pch=16)
```

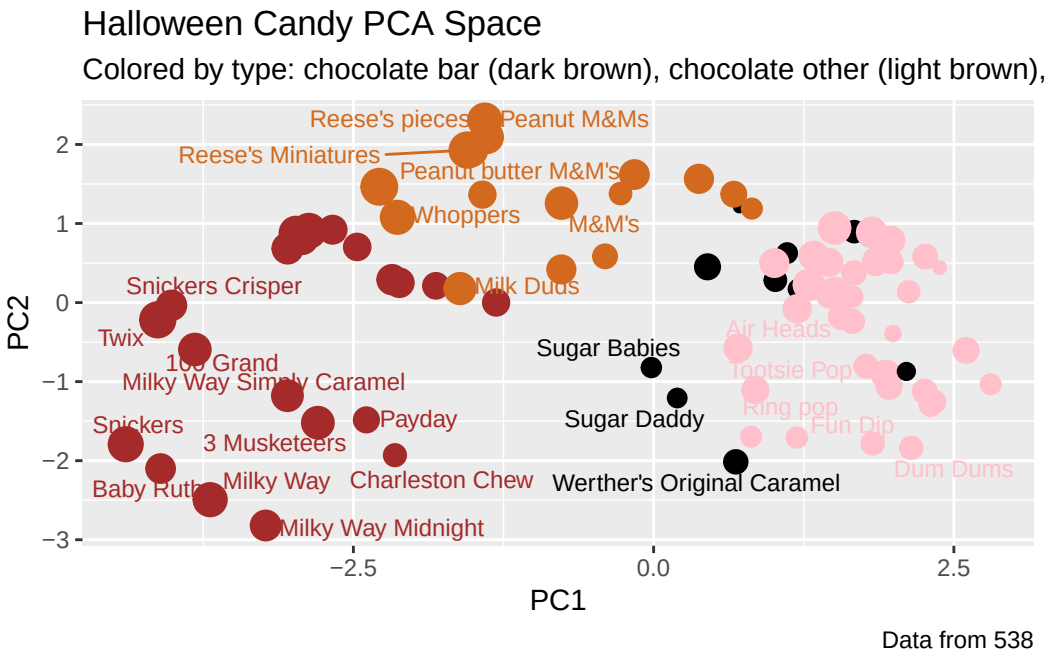


Plotting PCs with ggplot

```
my_data <- cbind(candy, pca$x[, 1:3]) # only need first 3 principal components

p <- ggplot(my_data) +
  aes(PC1, PC2, size=winpercent/100,
  text=rownames(my_data),
  label=rownames(my_data)) +
  geom_point(col = my_cols) +
  geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
    subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown), fru
    caption="Data from 538")
p
```

```
Warning: ggrepel: 59 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

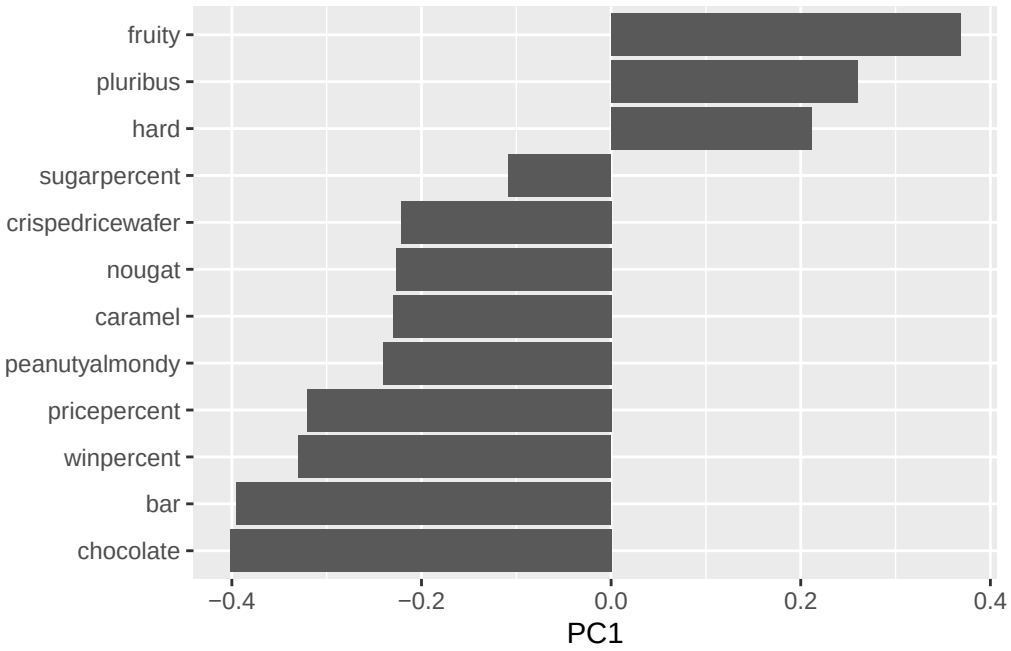


Using the `plotly` package to generate an interactive plot for html documents

```
#library(plotly)
#ggplotly(p) turned off for pdf output
```

Analyzing the PCA components

```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col() +
  labs(y = "")
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The original variables that are positively identified through PC1 were fruity, pluribus, and hard. This makes sense because a lot of fruity candies tend to be in hard candy formats and packaged in bundles. This was also seen in in the correlation matrix since the only positively correlated variables with fruity were hard and pluralists.

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A winning candies are typically chocolate, in a bar shape, and contain a peanut/nutty flavoring. From the visualization, it gives information about the general single categories that dominate the top win percentages such as how `chocolate` and `bar` typically have higher win percentages. From the correlation analysis, the top 3 highest correlated variables with win percent can be identified to further highlight specific variables that relate to win percent. Lastly from looking at PC1 components, `chocolate`, `bar`, and `peanutalmondy` are shown to have similar negative components as `winpercent` which further enforces how the presence of combinations of the 3 variables are related to the win percentages.